

ORCA Product Requirements Document (SIH26176)

ORCA — Product Requirements Document (PRD)

Marine EcOsystem Reasoning with Collaborative Agents

Field	Value
Project	ORCA — Marine EcOsystem Reasoning with Collaborative Agents
SIH Problem Statement	SIH26176
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0. Executive Summary

ORCA is an **agentic AI conversational platform** for marine intelligence. It understands natural-language marine questions (multilingual), decomposes them into tasks, retrieves **live and historical marine / geospatial / weather / satellite information from authoritative sources**, performs spatial-temporal reasoning, applies safety / geofencing / routing logic, and returns **explainable, evidence-backed recommendations** with maps, charts, alerts, and full source provenance.

0.1 Critical Product Principle

ORCA is NOT a static dataset repository. ORCA is a LIVE DATA INTELLIGENCE PLATFORM.

ORCA automatically acquires current information from official machine-readable sources whenever possible: REST APIs, ERDDAP, OGC/STAC services, official Python SDKs, authenticated data-download APIs, and feeds. **Web scraping is a last-resort connector**, used only when an authoritative source exposes no usable machine-readable interface, and always in compliance with robots.txt, terms of use, and rate limits.

The canonical statement, to be used in all documentation and demos:

"ORCA continuously acquires live data through the best available authoritative machine-readable interface, with API/service access first and compliant scraping only as a last resort."

The LLM never browses random websites and never ingests giant raw datasets. **The LLM calls ORCA data tools; the ORCA Data Gateway / Connector layer handles all source-specific communication.**

0.2 Core User Promise

A user can ask, in their own language:

- "I am near Kochi. Can I go fishing tomorrow morning?"
- "Where is the nearest favourable fishing zone?"
- "What are the wave and wind conditions near my fishing location?"
- "Which areas should I avoid because of cyclone, lightning, restricted waters or protected areas?"
- "Show me the safest route to the fishing zone."
- "Why is chlorophyll higher/lower than normal here?"
- "What are the current ocean conditions around this location?"
- "Is there an ISRO launch in the next two hours?"

For current or safety-critical questions, ORCA answers with **fresh, timestamped data** and shows both source-update and retrieval timestamps.

0.3 Non-Negotiable Requirements (Binding on all design decisions)

1. **API-first architecture.**
2. **ERDDAP is a first-class scientific data-access mechanism**, not a dataset.
3. Current/live/NRT/forecast data are strictly separated from historical datasets.

4. The existing **INCOIS Oceansat-2 OCM dataset (CHL, KD490, TSM; 2011-02-02 → 2020-05-01)** is **historical/reference** data, NOT current operational data.
5. **ERDDAP ≠ live data.** ERDDAP is the access layer; freshness depends on the dataset behind it.
6. For every source: record source update time, valid time, retrieval time, data type, latency, quality, provenance.
7. **Demand-driven live retrieval** for normal questions when cached data is stale.
8. **Forced refresh / live verification** for safety-critical or explicit "right now / current" questions.
9. Critical source checks run **in parallel** wherever possible.
10. Static datasets (EEZ / bathymetry / protected boundaries) are **locally indexed in PostGIS** and refreshed periodically — never fetched per user question.
11. **No secrets** in code, prompts, frontend bundles, Docker images, Git repositories or logs.
12. Environment variables / secret manager for all credentials.
13. **Never invent or guess an API endpoint.** Undocumented interfaces are labeled "*endpoint discovery/verification required.*"
14. Never bypass authentication, rate limits, robots restrictions, terms of use, or access controls.
15. Scraping only for authoritative sources with no machine-readable interface, obeying robots/terms/rate limits.
16. Every recommendation is traceable: source + dataset/service + variable + location + time + retrieval timestamp + processing step.
17. **Safety output is decision support, NOT official safety clearance or emergency authorization.**
18. **High chlorophyll does NOT guarantee fish.** PFZ is evidence/decision support presented with uncertainty.
19. If sources disagree materially, ORCA **shows the disagreement and lowers confidence** — it never silently selects one.
20. If required fresh data is unavailable or stale, ORCA **says so clearly.**

1. Product Scope

1.1 Primary Users

Persona	Needs	Primary Interface
Fishermen (artisanal & mechanized)	"Can I fish tomorrow? Where? Is it safe? Which route?"	Mobile chat + map, local language
Marine operators (harbour, vessel, charter)	Sea state, warnings, route risk, restricted waters	Desktop map + chat
Researchers	Historical CHL/SST trends, ERDDAP subsets, Argo/OBIS data	Desktop, data panels, export
Planners / coastal decision-makers	Warnings overview, zone status, freshness dashboard	Desktop dashboards

Persona	Needs	Primary Interface
Students / demo users	Guided demos, explainability of reasoning	Any device

1.2 Primary Use Cases (P0)

ID	Use Case	Acceptance Criterion (measurable)
UC-1	Fishing feasibility query ("Can I go fishing tomorrow morning near Kochi?")	End-to-end answer with PFZ candidates, sea-state, warnings, route, evidence panel in < 10 s live mode / < 3 s cached ; every value timestamped
UC-2	Nearest favourable fishing zone	Nearest non-rejected PFZ candidate returned with distance, bearing, risk class, and rejection reasons for closer-but-rejected candidates
UC-3	Current wave/wind conditions at a point	Values from ≥ 1 authoritative source with <code>source_updated_at</code> , <code>retrieved_at</code> , latency, and data-type label
UC-4	Hazard avoidance ("What should I avoid?")	All active warnings + restricted/protected polygons intersecting a configurable radius rendered on map and listed with provenance
UC-5	Safest route to a zone	Route avoiding geofenced polygons, depth constraints, and high-risk cells; risk score + explanation per segment
UC-6	ISRO launch status in next N hours	Critical-mode retrieval from official ISRO sources; answer shows mission, planned time, status if officially available, last-updated, source, confidence — or a clear "official live status unavailable" statement
UC-7	Chlorophyll anomaly explanation	Compares requested location/time against historical baseline (Oceansat-2 OCM 2011–2020, labeled HISTORICAL) and current ocean-colour source where available; explanation cites both

1.3 Secondary Use Cases (P1/P2)

- Fishing effort / vessel activity context (Global Fishing Watch, where licensed).
- Biodiversity occurrence context (OBIS).
- Subsurface validation (Argo).
- Historical climatology and trend analytics.
- Admin source-health monitoring and dataset registry management.

1.4 Out of Scope

- Official navigation certification or replacement of statutory navigation charts.
- Emergency/SAR dispatch or distress signaling (users are directed to official channels, e.g., 112 / Coast Guard 1554).
- Guaranteed fish-catch prediction.
- Vessel tracking of specific private vessels in real time (unless a specific licensed GFW dataset explicitly supports the claim).
- Autonomous vessel control.
- Offline-first operation (P0 requires connectivity; graceful degradation only).

1.5 Safety / Legal Disclaimers (shown in UI and answers)

1. *"ORCA provides decision support only. It is NOT an official safety clearance, weather warning authority, or emergency service."*
2. *"Always follow official IMD / INCOIS advisories and local harbour authority instructions. In emergencies call 112 or Indian Coast Guard 1554."*
3. *"Potential Fishing Zone information is scientific evidence with uncertainty. High chlorophyll does not guarantee fish presence."*
4. *"Data freshness depends on upstream providers. Every value shows its source and retrieval timestamps."*
5. Attribution and license notices per source (Section 38 / licensing appendix).

1.6 Multilingual Requirements

Initial languages: **English, Hindi, Malayalam, Tamil, Telugu, Kannada, Marathi, Bengali, Odia.**

Detailed behavior in Section 32.

1.7 Interaction Model

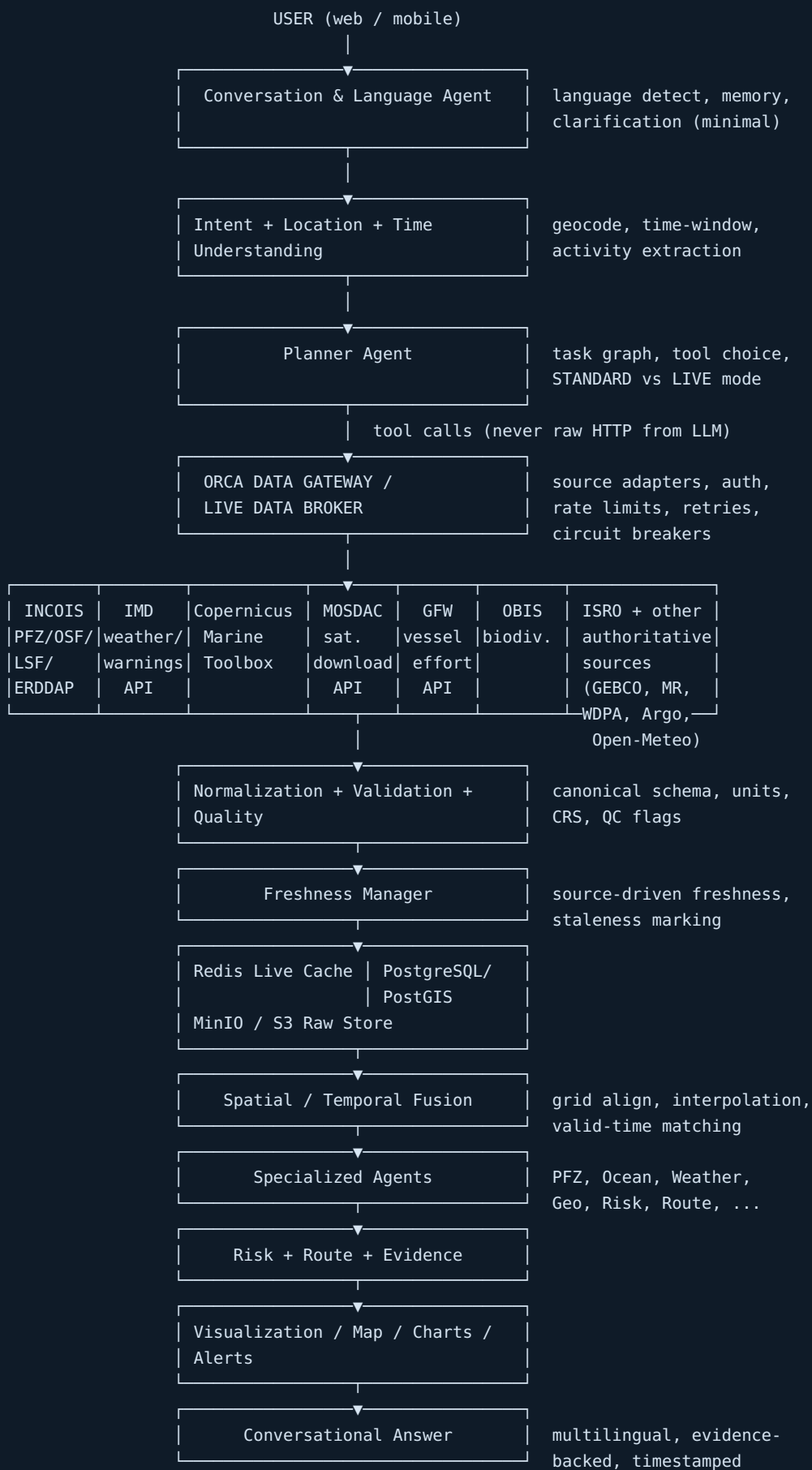
- **Chat-first + map-first:** the chat pane and interactive map are co-primary; every spatial answer is mirrored on the map.
- **Responsive:** mobile-first layout (fishermen on phones), full desktop layout for planners/researchers.
- **Live updates:** warnings and freshness badges update via WebSocket/SSE without page reload.

1.8 Global Acceptance Criteria

- Every numeric value in an answer carries: unit, `source`, `source_updated_at`, `retrieved_at`, data-type label (observation/forecast/NRT/historical/static), and confidence.
 - Every recommendation has an expandable **Evidence panel**.
 - Stale data is visibly badged ("STALE — last source update").
 - No response ever claims "live" without a verified source timestamp within the freshness policy window.
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2. High-Level Architecture

2.1 Baseline Architecture



2.2 Component Responsibilities, Interfaces, Failure Modes, Scaling

#	Component	Responsibility	Northbound interface	Southbound interface	Key failure modes	Failure handling	Scaling strategy
1	Conversation & Language Agent	Language detection, translation, memory, clarification	REST/WS from frontend	Planner (internal message)	LLM timeout, wrong language detect	Retry LLM, fall back to English + notice	Stateless; horizontal replicas
2	Intent/ Location/Time Understanding	NER for place/ time/activity; geocoding against local gazetteer + PostGIS	Called by Conversation Agent	PostGIS gazetteer	Ambiguous location	One clarification question max; else default radius around last known location	Stateless
3	Planner Agent	Task decomposition, tool selection, STANDARD vs LIVE mode decision	Conversation Agent	Data Gateway tool APIs, specialized agents	Bad plan, tool loop	Plan validation, max-step budget, timeout budget	Stateless
4	ORCA Data Gateway	Uniform Source Adapter execution: auth, rate-limit, retry, circuit breaker, provenance capture	Internal tool API (typed)	Provider APIs / services / SDKs	Provider down, auth failure, rate limited, schema drift	Section 18 fallback strategy; structured errors	Async worker pool; per-source concurrency caps
5	Normalization + Validation	Canonical schema (Sec. 11), unit/CRS/ time conversion, QC	Gateway output	Storage layer	Unit mismatch, malformed payload	Reject + quarantine raw payload in MinIO, alert	Stateless workers
6	Freshness Manager	Compute data age vs policy, mark stale, trigger refresh	All read paths	Redis metadata, source registry	Clock skew, missing source timestamp	Use retrieval time + conservative staleness; flag "source timestamp unavailable"	Library + Redis; trivial
7	Redis Live Cache	Hot values, freshness metadata,	Gateway/ agents	—	Eviction, restart	Cache-miss ⇒ refetch;	Redis cluster if needed

#	Component	Responsibility	Northbound interface	Southbound interface	Key failure modes	Failure handling	Scaling strategy
		rate-limit counters, circuit-breaker state				persistence optional	
8	PostgreSQL/ PostGIS	Operational + geospatial store (Sec. 27)	SQLAlchemy/ asyncpg	—	Disk, connection storm	PgBouncer, partitioning, read replica (scale)	Managed PG; partitions
9	MinIO/S3 Raw Store	Raw payloads: JSON/ NetCDF/ GeoTIFF/Zarr with provenance keys	Gateway, workers	—	Capacity	Lifecycle policies (Sec. 12)	Object storage scales natively
10	Spatial/ Temporal Fusion	Grid alignment, interpolation, valid-time matching (Sec. 16)	Agents	PostGIS + xarray on raw subsets	CRS bugs, extrapolation	Unit tests + uncertainty propagation; never extrapolate beyond forecast window	CPU workers (Celery)
11	Specialized Agents	Domain reasoning (Sec. 13)	Planner	Gateway tools, fusion services	Hallucination risk	Tool-only data access; schema-validated tool IO	Stateless
12	Risk + Route + Evidence	Hazard combination, route optimization, evidence assembly	Agents	PostGIS, fusion outputs	Over/under-conservative risk	Configurable policy thresholds; audit logs	CPU workers
13	Visualization layer	Map layers, charts, warning cards, source panel	Frontend	Backend layer APIs (GeoJSON/ tiles)	Large payloads	Vector tiles/ simplification; deck.gl for big layers	CDN + tile caching

2.3 Cross-Cutting Rules

- **The LLM only calls typed ORCA tools.** All external I/O flows through the Data Gateway.
- Every inter-component message carries a `request_id` for tracing (OpenTelemetry, Sec. 24).
- All southbound provider calls have bounded timeouts and are subject to per-provider rate limits and domain allowlists (SSRF protection, Sec. 23).

3. Live Data Acquisition Model

ORCA defines **four acquisition modes**, applied per source in strict preference order.

Mode A — Direct API / REST

Used when the provider exposes an official HTTP API. Examples: IMD API (documented at https://api.imd.gov.in/public/api_reference.html), Global Fishing Watch API, Protected Planet API v4, OBIS API, Open-Meteo.

Mode B — Scientific Data Service

Used for standards-based scientific access:

- **INCOIS ERDDAP** (<https://erddap.incois.gov.in/erddap/>)
- OGC APIs (WMS/WFS/WCS/OGC API-Features)
- STAC catalogs
- OPeNDAP endpoints
- other official data services

Mode C — Official Programmatic SDK / Data Client

Used when the provider's official access path is a client library or download API:

- **Copernicus Marine Toolbox Python API** (`copernicusmarine` package; docs: <https://help.marine.copernicus.eu/en/articles/7949409-copernicus-marine-toolbox-introduction> , subset API: <https://help.marine.copernicus.eu/en/articles/8283072-copernicus-marine-toolbox-api-subset>)
- **MOSDAC Data Download API/client** (manual: <https://www.mosdac.gov.in/downloadapi-manual> ; supports `datasetId` , `startTime` , `endTime` , `count` , `boundingBox` , download settings; requires MOSDAC account credentials)

Mode D — Scraping Fallback (LAST RESORT)

Used **only** when an authoritative source has no suitable machine-readable interface. Rules (full policy in Sec. 30):

- Prefer **network/backend endpoint discovery** over HTML scraping: if a webpage calls an official JSON/GeoJSON/NetCDF/WMS/WMTS service, use that service directly (after verifying it is publicly accessible and permitted).
- Obey robots.txt, terms of use, and rate limits; identify the ORCA user agent; never bypass CAPTCHAs/logins.

3.1 Source Adapter Interface

The rest of ORCA is **acquisition-mode agnostic**. Every source implements the `SourceAdapter` contract:

```
class SourceAdapter(Protocol):
    """Common conceptual interface. Adapters expose ONLY the methods
    their source actually supports. Unsupported methods raise
    CapabilityNotSupported – methods are never invented."""

    def get_metadata(self) -> SourceMetadata: ...           # registry info, capabilities
    def health_check(self) -> HealthStatus: ...             # reachability + auth + sample query
    def get_latest(self, q: SpatialQuery) -> Payload: ...    # most recent obs/NRT
    def get_forecast(self, q: SpatioTemporalQuery) -> Payload: ...
    def get_historical(self, q: SpatioTemporalQuery) -> Payload: ...
    def get_subset(self, q: SubsetQuery) -> Payload: ...    # var/time/lat/lon slicing
    def get_warnings(self, q: SpatialQuery) -> Payload: ... # alerts/advisories
    def get_status(self) -> ProviderStatus: ...            # provider-side status
```

Adapter guarantees (enforced by the Gateway base class): timeout, retry with exponential backoff + jitter, rate limiting, circuit breaker, provenance capture (raw payload → MinIO with request-id key), structured errors, metrics emission.

Capability matrix per adapter is declared in `get_metadata()` and stored in the Source Registry (Sec. 28) — the Planner can only select methods a source declares.

4. ORCA Data Gateway

The Data Gateway is the **single choke point** for all external data acquisition. No component — especially not the LLM — makes direct provider calls.

4.1 Gateway Responsibilities

1. Adapter registration and capability discovery (via Source Registry).
2. Credential injection from the secret manager (adapters never read env vars directly in production).
3. Per-provider rate limiting (token bucket, Redis-backed).
4. Per-provider circuit breaker (open on N consecutive failures; half-open probes).
5. Retry with exponential backoff + jitter (idempotent reads only).
6. Timeout budget enforcement (per-call and per-plan budget).
7. Provenance capture: raw payload → MinIO; provenance row → PostGIS (Sec. 29).
8. Normalization handoff to the canonical schema pipeline (Sec. 11).
9. Structured error taxonomy (Sec. 18) and metrics (Sec. 24).
10. Parallel fan-out execution for multi-source plans.

4.2 Required Adapters

Adapter	Acquisition Mode	Endpoint status	Supported methods (declared, not invented)
INCOIS PFZ Adapter	Verified INCOIS service/data endpoint	⚠ endpoint discovery/verification required	<code>get_metadata</code> , <code>health_check</code> , <code>get_latest</code> , <code>get_status</code>
INCOIS OSF Adapter	INCOIS operational service	⚠ endpoint discovery/verification required (Sec. 6)	<code>get_metadata</code> , <code>health_check</code> , <code>get_latest</code> , <code>get_forecast</code> , <code>get_status</code>
INCOIS LSF Adapter	INCOIS service/backend	⚠ endpoint discovery/verification required (Sec. 6)	<code>get_metadata</code> , <code>health_check</code> , <code>get_forecast</code> , <code>get_status</code>
INCOIS ERDDAP Adapter	Mode B — ERDDAP	✅ base URL documented: https://erddap.incois.gov.in/erddap/ ; per-dataset endpoints must be validated (Sec. 5)	<code>get_metadata</code> , <code>health_check</code> , <code>get_historical</code> , <code>get_subset</code> , <code>get_latest</code> * (*only for datasets validated as current/NRT)
IMD Weather Adapter	Mode A — official REST API	✅ API reference documented: https://api.imd.gov.in/public/api_reference.html ; per-endpoint contracts to be confirmed against the reference	<code>get_metadata</code> , <code>health_check</code> , <code>get_latest</code> , <code>get_forecast</code> , <code>get_status</code>
IMD Marine Warning Adapter	Mode A — official REST API where available	⚠ verify warning endpoints in IMD API reference	<code>get_metadata</code> , <code>health_check</code> , <code>get_warnings</code> , <code>get_status</code>
Copernicus Ocean Physics Adapter	Mode C — Copernicus Marine Toolbox	✅ official Python SDK documented	<code>get_metadata</code> , <code>health_check</code> , <code>get_latest</code> , <code>get_forecast</code> , <code>get_historical</code> , <code>get_subset</code>
Copernicus Ocean Colour Adapter	Mode C — Toolbox	✅ official SDK	<code>get_metadata</code> , <code>health_check</code> , <code>get_latest</code> , <code>get_historical</code> , <code>get_subset</code>
Copernicus Waves Adapter	Mode C — Toolbox	✅ official SDK	<code>get_metadata</code> , <code>health_check</code> , <code>get_latest</code> , <code>get_forecast</code> , <code>get_subset</code>
MOSDAC Satellite Adapter	Mode C — MOSDAC Download API	✅ API manual documented: https://www.mosdac.gov.in/downloadapi-manual ; requires MOSDAC credentials	<code>get_metadata</code> , <code>health_check</code> , <code>get_latest</code> , <code>get_historical</code> , <code>get_subset</code>
Global Fishing Watch Adapter	Mode A — official REST API	✅ documented: https://globalfishingwatch.org/our-apis/documentation/ ; requires API token	<code>get_metadata</code> , <code>health_check</code> , <code>get_latest</code> †,

Adapter	Acquisition Mode	Endpoint status	Supported methods (declared, not invented)
			<code>get_historical</code> , <code>get_subset</code> († latency-tracked; never claimed as exact real-time positions unless the specific dataset supports it)
OBIS Adapter	Mode A — API (small queries); GeoParquet/ AWS (large subsets)	✓ documented: https://portal.obis.org/data/access/	<code>get_metadata</code> , <code>health_check</code> , <code>get_historical</code> , <code>get_subset</code>
GEBCO Adapter	Regional download / OPeNDAP / data service	✓ GEBCO_2026 grid page: https://www.gebco.net/data-products-gridded-bathymetry-data/gebco2026-grid	<code>get_metadata</code> , <code>health_check</code> , <code>get_subset</code> (static; ingested to PostGIS/ raster store)
Marine Regions Adapter	Download + local PostGIS (+ OGC services where useful)	✓ downloads page: https://www.marineregions.org/downloads.php ; World EEZ v12 listed	<code>get_metadata</code> , <code>health_check</code> , <code>get_subset</code> (static, versioned)
Protected Planet / WDPA Adapter	Mode A — official API v4 or GIS download	✓ documented: https://api.protectedplanet.net/documentation ; token required; ⚠ license/non-commercial restrictions must be checked before any deployment	<code>get_metadata</code> , <code>health_check</code> , <code>get_subset</code>
Argo Adapter	GDAC / AWS / programmatic access	✓ access options documented: https://www.argodatamgt.org/DataAccess.html	<code>get_metadata</code> , <code>health_check</code> , <code>get_historical</code> , <code>get_subset</code>
ISRO Mission Information Adapter	Official ISRO web/data sources; API availability varies	⚠ NO invented API; source monitoring + verified machine-readable endpoints only (https://www.isro.gov.in/)	<code>get_metadata</code> , <code>health_check</code> , <code>get_status</code> , <code>get_latest</code> ‡ (‡ only if a verified machine-readable source exists; otherwise compliant monitored web source with explicit limitation statement)
Open-Meteo Supplementary Weather Adapter	Mode A — public API	✓ https://open-meteo.com/ ; supplementary ONLY — never replaces IMD/INCOIS warnings; verify current terms/rate limits before production	<code>get_metadata</code> , <code>health_check</code> , <code>get_latest</code> , <code>get_forecast</code>

4.3 Adapter Development Rules

- An adapter may only expose methods its source genuinely supports; everything else raises `CapabilityNotSupported` .

- Every adapter ships with: contract tests against recorded fixtures, a mock implementation for local dev, and a health-check sample query.
- Adapters marked "*endpoint discovery/verification required*" are implemented in two phases:
 1. **Discovery phase** — inspect official interface network calls, confirm public accessibility and permission, document the contract in the Source Registry, set `endpoint_verified = true`.
 2. **Implementation phase** — build the connector against the verified contract only.
- The Gateway refuses to route production traffic to any adapter with `endpoint_verified = false` (mock/stub data clearly labeled in non-production).

5. ERDDAP — Detailed Requirement

ERDDAP is one of the most important ORCA components and must be understood precisely:

- **ERDDAP is a data-access server/system, not a dataset.**
- It allows programmatic queries against datasets hosted by the provider.
- ORCA queries **only the required variables, time, latitude and longitude** instead of downloading an entire dataset.
- ERDDAP can return machine-readable formats such as **CSV/JSON/NetCDF**, depending on dataset/server configuration.
- **Freshness depends on the dataset behind ERDDAP** — an ERDDAP query is not evidence of live data.

5.1 Official Server

- INCOIS ERDDAP: <https://erddap.incois.gov.in/erddap/>
- Server status page: <https://erddap.incois.gov.in/erddap/status.html>

5.2 Known Historical Dataset (classified HISTORICAL)

Field	Value
Dataset ID	<code>incois_oceansat2_datasets</code>
Variables	CHL — chlorophyll-a; KD490 — diffuse attenuation coefficient; TSM — total suspended matter
Temporal coverage	2011-02-02 → 2020-05-01
Classification	HISTORICAL / reference — never presented as current operational ocean condition
Use	Climatology baseline, anomaly explanation (UC-7), research/demo of ERDDAP subsetting

5.3 ERDDAP Dataset Registry

ORCA maintains an **ERDDAP Dataset Registry** (table `datasets` , Sec. 27) storing for every registered dataset:

`dataset_id` , `title` , `provider` , `variable list` , `units` , `spatial coverage` , `temporal coverage` , `last available source time` , `last successful retrieval` , `update frequency` , `freshness class` , `live/NRT/historical flag` , `access URL` , `response format` , `authentication requirement` , `quality metadata` , `status` .

5.4 ERDDAP Connector Capabilities

The connector MUST support: variable selection · time slicing · latitude slicing · longitude slicing · stride/downsampling where appropriate · response-format selection · retries · timeout · caching · provenance · validation · missing-value handling.

Conceptual query pattern (pattern only — developers MUST validate the exact dataset endpoint against the live INCOIS ERDDAP catalog before implementation; no fabricated URLs are to be committed to configuration):

```
GET {ERDDAP_BASE}/griddap/{dataset_id}.{format}?{variable}
    [(time_start):stride:(time_end)]
    [(lat_min):stride:(lat_max)]
    [(lon_min):stride:(lon_max)]

dataset + variable + time + lat + lon → small subset → normalize → cache
```

5.5 Discovery Process for CURRENT/NRT INCOIS ERDDAP Datasets

A dataset may only be registered as current/NRT after this validated process:

1. Query/list the current ERDDAP catalog (e.g., `allDatasets` / `info` services).
2. Inspect dataset metadata (variables, units, coverage).
3. Identify the current temporal end date.
4. Inspect last update information.
5. Verify operational/NRT status with the provider documentation.
6. Test a small subset query and validate the response.
7. **Register the dataset only after validation** (sets `endpoint_verified = true` , freshness class).
8. Automatically monitor freshness thereafter (background worker, Sec. 21).

Datasets failing any step remain unregistered or flagged `status = 'unverified'` and are never used in answers.

6. INCOIS Operational Services

Official INCOIS operational systems are the primary Indian source for current/forecast marine information.

6.1 Ocean State Forecast (OSF)

- Official interface: <https://www.incois.gov.in/oceanservices/osfforecast.jsp>
- Known operational variables: wind speed/direction · surface current speed/direction · significant wave height · swell height · wave period · swell period · sea surface temperature · mixed layer depth · D20 (depth of 20 °C isotherm) · additional ocean/satellite products shown by the service.

6.2 Location Specific Forecast (LSF)

- Official interface: <https://incois.gov.in/oceanservices/LSF/index.html>
- Location-specific wave/current forecasts.

6.3 Interface Classification (mandatory distinction)

The PRD and implementation strictly distinguish, for every INCOIS capability:

Class	Meaning	Example
Documented public API	Officially published machine contract	(none currently confirmed for OSF/LSF)
Operational web interface	Human-facing page	OSF/LSF pages above
Underlying service/ backend	Machine endpoints the pages call	⚠ to be discovered/verified
ERDDAP dataset	Dataset served via INCOIS ERDDAP	<code>incois_oceansat2_datasets</code> (historical)
Downloadable product	Files offered for download	per INCOIS Data Holdings (https://incois.gov.in/site/dataholdings.jsp)

It is **FORBIDDEN** to invent a REST endpoint such as `/forecast?lat=...` if it is not officially documented.

6.4 OSF/LSF Connector Plan (endpoint discovery/verification required)

1. Investigate the actual network requests made by the official OSF/LSF interfaces (browser network inspection).
2. Identify machine-readable endpoints/services (JSON/GeoJSON/WMS/etc.) **if publicly accessible and permitted**.
3. Prefer the underlying official service over HTML scraping.

4. Create the connector **only after endpoint/contract verification**, recording the contract in the Source Registry.
5. If no public machine interface is available or permitted, define a compliant fallback strategy (Sec. 30 hierarchy: official downloadable products → compliant, rate-limited HTML parsing of the official page with provenance and breakage monitoring) and/or contact INCOIS for authorized access.

7. Source Matrix

Legend — Freshness: LIVE (minutes) / NRT (hours) / FC (forecast) / HIST (historical) / STATIC.
 Priority: P0 = MVP-critical, P1 = enhanced, P2 = advanced.

#	Source	Purpose	Best access method	Type	Credential required?	Expected latency
1	INCOIS PFZ	PFZ / fishing advisory	INCOIS service/data endpoint — endpoint must be verified	Advisory (issued cadence)	Unknown until endpoint verification	Advisory cadence (typ. daily)
2	INCOIS OSF	Waves, wind, currents, SST forecasts	INCOIS operational service — endpoint must be verified	FC	Unknown until endpoint verification	Forecast cycle (typ. 6–12 h)
3	INCOIS LSF	Location-specific wave/ current forecast	INCOIS service/backend after verification	FC	Unknown until endpoint verification	Forecast cycle
4	INCOIS OCM (Oceansat-2)	CHL/KD490/ TSM reference	ERDDAP (incois_oceansat2_datasets)	HIST (2011-02-02 → 2020-05-01)	Public ERDDAP access expected — verify per dataset	Seconds (subset query)
5	IMD weather	Weather obs/ forecast	Official REST API — https://api.imd.gov.in/public/api_reference.html	LIVE/FC	Verify: API key may be required per endpoint	Minutes
6	IMD marine warnings	Cyclone/sea-area warnings	Official REST API where available	LIVE (warning)	Verify per endpoint	1–5 min target

#	Source	Purpose	Best access method	Type	Credential required?	Expected latency
7	Copernicus Ocean Physics	Currents, SST, salinity, MLD	Copernicus Marine Toolbox Python API	NRT/FC/HIST	Copernicus account (username/password)	Product cadence (hours)
8	Copernicus Ocean Colour	CHL current/ NRT	Toolbox/API	NRT	Copernicus account	Daily-ish product cadence
9	Copernicus Waves	Wave forecast/ NRT	Toolbox/API	NRT/FC	Copernicus account	Product cadence
10	MOSDAC	ISRO satellite products	Official MOSDAC Download API (https://www.mosdac.gov.in/downloadapi-manual) — supports datasetId, startTime, endTime, count, boundingBox	NRT/HIST	MOSDAC account required — never hardcoded	Product-dependent
11	Global Fishing Watch	Fishing effort/ vessel activity	Official REST API (https://globalfishingwatch.org/our-apis/documentation/)	Derived/modelled, latency-tracked	API token required	Dataset-dependent (days typical) — tracked; never described as exact real-time positions unless the specific dataset supports that claim
12	OBIS	Biodiversity occurrences	API (small queries); GeoParquet/ AWS (large) — https://portal.obis.org/data/access/	HIST (occurrences)	None expected — verify chosen endpoint	Seconds–minutes
13	GEBCO	Bathymetry	Regional download / OPeNDAP / data service — GEBCO_2026 grid (https://www.gebco.net/data-products-gridded-bathymetry-data/gebco2026-grid)	STATIC (versioned)	None	Local (pre-ingested)

#	Source	Purpose	Best access method	Type	Credential required?	Expected latency
14	Marine Regions	EEZ, maritime boundaries	Download + local PostGIS (+ OGC services where useful) — https://www.marineregions.org/downloads.php	STATIC (versioned; World EEZ v12 currently listed — not "live"; version and periodically check for updates)	Registration for some downloads — verify	Local
15	Protected Planet / WDPA	Protected areas	Official API v4 (https://api.protectedplanet.net/documentation) or GIS download	STATIC/monthly updates	API token required;  check license/ non-commercial restrictions before any deployment	Local after ingest
16	Argo	Subsurface T/ S profiles	GDAC/AWS/programmatic (https://www.argodatamgt.org/DataAccess.html)	NRT/HIST	None	Hours (NRT stream)
17	ISRO mission info	Launch schedule/ status	Official ISRO web/data sources (https://www.isro.gov.in/); API availability varies — do NOT invent an API ; source monitoring and/or verified machine-readable endpoints	Event/status	Unknown until endpoint verification	1–5 min monitoring in critical mode
18	Open-Meteo	Supplementary weather/model	Public API (https://open-meteo.com/)	FC/NRT model	None for free tier — no secret embedded unless the chosen tier explicitly requires one; check current terms/rate limits before production	Seconds

7.1 Matrix Governance

- Every row maps to a Source Registry record (Sec. 28) with `endpoint_verified` status.
- Rows with "endpoint must be verified" are **discovery-phase work items** in the P0 backlog — implementation is blocked until verification.

- Latency and freshness columns are engineering targets, **not provider guarantees**.

8. Credentials / API Keys / Secrets

This PRD NEVER contains real secret values. All values below are placeholders / environment-variable names.

8.1 Credential Inventory

Source / Service	Credential Required?	Env variable(s)	Notes
Application core	—	APP_ENV , APP_SECRET_KEY , JWT_SECRET_KEY	App/JWT secrets from secret manager
PostgreSQL/ PostGIS	account	DATABASE_URL , POSTGIS_URL	Connection strings without inline plaintext in logs
Redis	optional auth	REDIS_URL	
MinIO/S3	keys	MINIO_ENDPOINT , MINIO_ACCESS_KEY , MINIO_SECRET_KEY	
LLM provider	token required	OPENAI_API_KEY (or LLM_PROVIDER_API_KEY if another provider is selected)	Backend-only
INCOIS services	unknown until endpoint verification	INCOIS_API_KEY , INCOIS_USERNAME , INCOIS_PASSWORD	Only include variables actually required by verified INCOIS services; public ERDDAP access may not require credentials — verify per dataset
IMD API	verify per endpoint	IMD_API_KEY , IMD_USERNAME , IMD_PASSWORD	Only if the verified IMD endpoint requires credentials
Copernicus Marine	account required	COPERNICUSMARINE_SERVICE_USERNAME , COPERNICUSMARINE_SERVICE_PASSWORD	Free registration
MOSDAC	account required	MOSDAC_USERNAME , MOSDAC_PASSWORD	Required for downloads; never hardcoded
Global Fishing Watch	token required	GFW_API_TOKEN	Terms compliance required
Protected Planet	token required	PROTECTED_PLANET_API_TOKEN	License review before deployment
OBIS		OBIS_API_KEY	Only if the selected OBIS endpoint requires it

Source / Service	Credential Required?	Env variable(s)	Notes
	unknown until endpoint verification		
ISRO-related services	unknown until endpoint verification	ISRO_API_KEY	Only if a verified service requires it
Map provider	service-specific	MAPBOX_ACCESS_TOKEN	OR use MapLibre with a permitted tile provider and no proprietary key (preferred)
Open-Meteo	no credential (free tier)	—	Unless the chosen service tier explicitly requires one
Monitoring	optional	SENTRY_DSN	

8.2 Secret-Handling Policy (mandatory)

1. **Production:** all credentials live in a **secrets manager** (e.g., Vault / cloud secret manager / sealed K8s secrets).
2. **.env is allowed only for local development;** `.env` is git-ignored; only `.env.example` (placeholders, Sec. 37) is committed.
3. **The frontend never receives server-side API keys.** All authenticated third-party calls go through the backend proxy/Gateway.
4. Credentials are **rotated** on schedule and on suspicion of exposure.
5. Credentials are **redacted from logs**, traces, error reports, and provenance records (query parameters are stored *excluding secrets*).
6. Outbound calls are **rate-limited** per provider.
7. Third-party tokens are **never exposed in browser code**, bundles, or source maps.
8. CI secret-scanning (e.g., gitleaks) blocks any commit containing secret patterns.

9. Live Data / Freshness Engine

9.1 Data-Item Metadata (mandatory on every stored/returned item)

Every data item carries:

source · dataset/service · variable · value · unit · latitude · longitude ·
observed_at · valid_from · valid_until · source_updated_at · retrieved_at ·
data_type (observation / NRT / forecast / historical / static) · quality_flag ·
processing_version · provenance URL · confidence .

9.2 Freshness Classes (engineering targets — NOT provider guarantees)

Class	Data	Cache target / behavior
CRITICAL	Warnings/alerts	1–5 min cache target; forced refresh for critical queries
CRITICAL	Launch/event status	1–5 min monitoring when the user asks current status
HIGH	Weather / current marine conditions	5–15 min target
HIGH	Waves/wind/current forecast	15–60 min target
MEDIUM	SST / ocean colour	Hours-scale, based on actual source update cadence
LOW	PFZ	Refresh when a new advisory is issued / provider cadence
STATIC	EEZ / bathymetry / protected areas	Versioned periodic refresh (background worker)

9.3 Source-Driven Freshness Algorithm

```
on read(variable, location, time):
    item = cache.get(key)
    policy = freshness_policy(source, variable, query_mode)
    if item exists:
        age_source = now - item.source_updated_at    # authoritative age
        age_retrieval = now - item.retrieved_at
        if item.source_updated_at is missing:
            treat conservatively: age = age_retrieval; flag "source timestamp unavailable"
        if age within policy AND query_mode == STANDARD:
            return item (with freshness badge)
    # stale, missing, or LIVE mode:
    fresh = gateway.fetch(source, ...)                # parallel across sources
    latency = fresh.retrieved_at - fresh.source_updated_at
    if fresh.source_updated_at older than policy.max_source_age:
        mark item STALE-AT-SOURCE; surface prominently in answer
    cache.put(fresh); return fresh
```

Rules:

- **Never claim "live" solely because a connector was called** — inspect the source timestamp, calculate latency, mark stale data.
- Freshness badges and both timestamps are **always shown in the UI**.
- Freshness metrics per source feed the health dashboard (Sec. 24).

10. Critical Query Mode

10.1 STANDARD Mode

- Use cached data if within the freshness policy.
- Fetch from source only when data is missing or stale.

- Parallelize independent source requests.

10.2 LIVE / CRITICAL Mode

Triggers (intent classifier + keyword/temporal heuristics): "right now" · "current" · "today" · next 1–3 hours windows · "safe/danger" · "fishing now" · "cyclone" · "lightning" · "severe weather" · "emergency" · "launch in next N hours" · explicit user request for fresh data.

Behavior:

1. Bypass stale cache entirely.
2. Force refresh of the authoritative sources for the question.
3. Verify source timestamps on every returned item.
4. Cross-check multiple sources where possible (Sec. 17).
5. Lower confidence when sources disagree.
6. **Never provide a safety guarantee** — output remains decision support with disclaimers.
7. If a required authoritative source cannot deliver fresh data, the answer states this explicitly and labels any fallback values.

10.3 Mode Selection Contract

The Planner records `query_mode` in the plan; the Gateway enforces it (`force_refresh=true` per call). Mode, per-source refresh results, and timestamp checks appear in the Evidence panel.

11. Data Normalization

11.1 Canonical Data Schema

```
{
  "source": "incois_osf",
  "dataset": "osf_regional_forecast",
  "variable": "significant_wave_height",
  "value": 1.42,
  "unit": "m",
  "latitude": 9.9312,
  "longitude": 76.2673,
  "observed_at": "2026-09-10T06:00:00Z",
  "valid_time": "2026-09-11T03:00:00Z",
  "retrieved_at": "2026-09-10T09:14:22Z",
  "data_type": "forecast",
  "quality_flag": "good",
  "source_latency_seconds": 123,
  "confidence": 0.92,
  "provenance_url": "s3://orca-raw/raw/incois/2026/09/10/req-8f3a.json"
}
```

11.2 Normalization Rules

Dimension	Rule
Units	SI-first canonical units (m, m/s, °C, mg/m³ ...); conversions via a registry (pint); original unit preserved in provenance
Timestamps / time zones	Everything stored in UTC ISO-8601 ; UI localizes to IST/user locale; source-local times converted at ingest
Longitude conventions	Canonical -180...180 ; 0...360 sources converted at ingest
Missing values	Provider sentinels (-999, NaN encodings, <code>_FillValue</code>) mapped to explicit nulls with <code>quality_flag = "missing"</code>
Quality flags	Provider QC flags mapped to canonical set: <code>good</code> / <code>suspect</code> / <code>bad</code> / <code>missing</code> / <code>unknown</code> (original preserved)
CRS	Canonical EPSG:4326 for exchange; projected CRS (e.g., UTM 43N/44N) only inside metric computations; pyproj-based transforms
Forecast vs observation	<code>data_type</code> mandatory; forecast items must carry <code>valid_time</code> distinct from issue time
Source version	Product/model version recorded per item (<code>processing_version</code> , source version in provenance)

Validation failures quarantine the raw payload (MinIO), emit `normalization_error` metric, and never reach the answer path.

12. Storage

12.1 Storage Tiers

Tier	Technology	Contents
Hot / live cache	Redis	Latest values, freshness metadata, rate-limit counters, circuit-breaker state, session/task state
Operational + geospatial	PostgreSQL + PostGIS	Canonical observations/forecasts, warnings, PFZ zones, boundaries, protected areas, routes, registry, evidence, users (Sec. 27)
Raw / large files	MinIO / S3	Raw API responses, NetCDF, GeoTIFF, Zarr, large files — keyed by provenance path (Sec. 29)

12.2 Storage Policies

- **Object-store lifecycle policies** on raw data (e.g., raw JSON 90 days → archive tier; NetCDF per dataset policy).

- Database **indexes on time / location / source**; **PostGIS GiST/SP-GiST** indexes on all geometry columns.
- **Partition large time-series tables** (`data_observations` , `forecasts`) by time and/or source when volume requires.
- **Never put raw 210 MB+ NetCDF files into the LLM context.** Scientific files are subset server-side (xarray) and only small, normalized extracts reach agents.

12.3 Scientific Processing Stack

`xarray` · `NumPy` · `pandas` · `GeoPandas` · `Shapely` · `rasterio` · `netCDF4` / `h5py` · `Zarr` · `PyArrow` / `Parquet`.

13. Agent Architecture

All agents access data **exclusively through typed Gateway tools**. Tool inputs/outputs are Pydantic-validated; agents cannot emit free-form external calls.

#	Agent	Responsibilities	Tools used	Guardrails
1	Conversation & Language Agent	Language detection · multilingual response · conversation memory · query clarification only when essential	translation/LLM, session store	Max 1 clarification per turn; never blocks safety answers on clarification
2	Planner Agent	Decomposes request · chooses required tools · determines time/ location · selects STANDARD vs LIVE mode	tool catalog, source registry	Plan schema validation; step & time budget; no unverified adapters
3	Marine Data Discovery Agent	Finds appropriate source/dataset · checks source registry · checks freshness	registry query, freshness API	Only registered + verified datasets
4	PFZ / Fishing Intelligence Agent	PFZ retrieval · fishing suitability evidence · CHL/SST interpretation	PFZ tool, ocean-colour tools, historical baseline	Never equates CHL directly with guaranteed fish presence ; outputs uncertainty
5	Ocean Intelligence Agent	Waves · currents · SST · salinity/ MLD when relevant · ocean trends	OSF/LSF, Copernicus, ERDDAP tools	Labels forecast vs observation vs NRT
6	Weather Intelligence Agent	IMD · wind · rain · cyclone · lightning · marine warnings	IMD tools, Open-Meteo (supplement)	Authoritative warnings can never be overridden by supplementary sources
7	Geospatial Agent	EEZ · territorial waters · protected areas · restricted zones · depth · distance · polygon intersection · geofencing	PostGIS spatial tools	Uses locally indexed static layers; versions cited

#	Agent	Responsibilities	Tools used	Guardrails
8	Risk Assessment Agent	Combines hazards · classifies risk · confidence · source agreement · rejects unsafe candidates	fusion outputs, warning state	Conservative default thresholds; policy-configurable; audit-logged decisions
9	Route Optimization Agent	Distance · wave/wind/current penalties · geofence constraints · depth constraints · risk penalties	PostGIS routing grid, risk surface	Route must not intersect rejected polygons (hard constraint)
10	Analytics / Biodiversity Agent	OBIS · Argo · historical trends · biodiversity only where relevant	OBIS/Argo/ERDDAP historical tools	Historical labels enforced
11	Evidence / Explanation Agent	Source citations · timestamps · values · processing explanation · confidence · disagreements	provenance store	Every claim maps to ≥ 1 evidence record
12	Visualization / Reporting Agent	Map layers · charts · route · warning cards · source panel	layer/chart APIs	Payload size limits; simplification

13.1 Risk Scoring Model (Risk Assessment Agent)

```

risk(zone, window) = weighted_max(
    sea_state_risk(SWH, swell, period, wind),
    warning_risk(active_warnings n zone),
    geofence_risk(zone n restricted/protected),
    depth_risk(bathymetry vs vessel draft class),
    route_exposure_risk(path hazards)
) × confidence_adjustment(source_agreement, freshness)

classes: SAFE / CAUTION / UNSAFE / NO-GO (warning override)

```

- Any active authoritative warning intersecting a zone ⇒ **NO-GO** regardless of other factors (AC-D).
- Thresholds (e.g., SWH limits per vessel class) are configuration, documented in the ops runbook, never hardcoded magic numbers.

14. Main Demo Workflow (Flagship)

USER: "I am near Kochi. Can I go fishing tomorrow morning, and where is the nearest favourable fishing zone?"

Step	Actor	Action
1	Conversation Agent	Extracts: location = Kochi (9.9312 N, 76.2673 E ± radius), date = tomorrow, time window = morning (e.g., 05:00–11:00 IST), activity = fishing
2	Planner	

Step	Actor	Action
		Creates tasks: PFZ · waves · wind · currents · SST · weather · marine warnings · lightning/cyclone · geofencing · depth · route
3	Data Gateway	Checks freshness for each required item
4	Gateway (parallel fan-out)	INCOIS PFZ · INCOIS OSF/LSF · IMD · Copernicus (if required) · local PostGIS for boundaries/depth
5	Normalization	Canonical schema, units, UTC, QC
6	Fusion	Spatial alignment (grid/point harmonization around Kochi sector)
7	Fusion	Temporal alignment (forecast valid times $\cap$ tomorrow-morning window)
8	PFZ Agent	Generates candidate PFZs
9	Risk + Geospatial Agents	Remove: unsafe sea states · warning zones · restricted zones · protected zones (if applicable) · shallow/depth-incompatible zones · excessive route risk
10	Risk Agent	Ranks remaining candidates (distance, risk, confidence)
11	Route Agent	Computes safest feasible route to the top candidate
12	Evidence Agent	Explains: why the candidate was selected · why alternatives were rejected · source · timestamps · confidence
13	UI	Shows: map · candidate zones · route · hazard overlays · source evidence · timestamps · risk score

Demo acceptance: all 13 steps traceable in the Evidence panel; end-to-end latency within Sec. 33 targets; rejection reasons visible for every discarded candidate.

15. Second Demo: Live ISRO Launch Information

USER: "Is ISRO launching a rocket in the next two hours?"

Flow:

1. Classify as **event/current** query → enter **CRITICAL/LIVE mode**.
2. Query official ISRO sources via the ISRO Mission Information Adapter.
3. Retrieve the latest mission schedule/status from a **verified machine-readable or official web source** (per Sec. 30 hierarchy; never an invented API).
4. Compare timestamps (page/feed update time vs now).
5. **If official live status is unavailable, state that limitation explicitly** — e.g., "Official ISRO sources list mission X planned for (last updated). A real-time launch status feed is not publicly available; confidence: medium."
6. Answer shows: **mission · planned time · current status (if officially available) · last updated · source · confidence**.

16. Spatial / Temporal Reasoning

Required capabilities (implemented in PostGIS + Python fusion services):

Capability	Implementation note
Nearest-neighbor search	PostGIS <-> KNN with GiST index
Point-in-polygon	ST_Contains / ST_Intersects on indexed layers
Polygon intersection	ST_Intersection , validity-checked geometries
Buffer	Geodesic buffers via geography type / projected CRS
Distance	ST_DistanceSphere / geography distance (meters)
Route intersection	Route linestring vs hazard/geofence polygons
Temporal overlap	Interval algebra on valid_from/valid_until
Forecast valid-time matching	Select forecast step(s) covering the requested window; interpolate between steps only within model bounds
Spatial interpolation	Bilinear/nearest on grids only where scientifically justified ; method recorded in provenance
Grid alignment	xarray regridding to a common analysis grid per query sector
Coordinate transformation	pyproj; canonical EPSG:4326 exchange
Uncertainty propagation	Where feasible: combine source confidence, interpolation distance penalty, temporal distance penalty into item confidence

17. Confidence / Source Agreement

When multiple sources provide the same variable:

1. Compare values on the aligned grid/point.
2. Calculate spread (e.g., relative range or coefficient of variation).
3. Identify disagreement beyond a per-variable threshold.
4. Weight sources by **authority and suitability** (authoritative Indian operational sources rank first for Indian waters).
5. **Never silently average incompatible measurements** (different depths, valid times, or data types are not averaged).

Examples:

- INCOIS wind = 14 kt, IMD = 16 kt, Copernicus = 15 kt → **high agreement**, high confidence.

- INCOIS = 12 kt, IMD = 35 kt, Copernicus = 14 kt → **low agreement** → investigate (valid-time mismatch? unit error?) or report uncertainty explicitly and lower confidence; show all values in Evidence.

Confidence is a function of: source authority · freshness · spatial distance · temporal distance · quality flags · agreement · forecast-vs-observation · missingness.

```
confidence = base_authority
    × freshness_factor(age vs policy)
    × spatial_factor(distance to nearest sample)
    × temporal_factor(|t_query - valid_time|)
    × quality_factor(qc_flag)
    × agreement_factor(spread)
    × type_factor(observation > NRT > forecast > historical proxy)
    × completeness_factor(missingness)
```

18. API Failure / Fallback Strategy

18.1 Mandatory Connector Behaviors

Every connector supports: **timeout** (per-call budget) · **retry with exponential backoff + jitter** (idempotent reads; capped attempts) · **rate-limit handling** (respect `Retry-After` ; local token bucket) · **circuit breaker** (open after N failures; half-open probes) · **health check** · **fallback source** (declared in registry) · **stale-cache fallback with explicit warning** · **structured error** · **logging** · **metrics**.

18.2 Error Taxonomy

Code	Meaning	Handling
<code>TIMEOUT</code>	Call exceeded budget	Retry (bounded) → fallback
<code>AUTH_FAILED</code>	Credential rejected	No retry; alert ops; fallback
<code>RATE_LIMITED</code>	429/provider limit	Backoff per <code>Retry-After</code> ; queue
<code>PROVIDER_DOWN</code>	5xx/unreachable	Circuit breaker; fallback
<code>SCHEMA_DRIFT</code>	Response failed contract validation	Quarantine payload; alert; fallback
<code>STALE_AT_SOURCE</code>	Source data older than policy	Serve with prominent staleness notice
<code>NOT_FOUND</code>	No data for query	Partial result + explanation
<code>CAPABILITY_UNSUPPORTED</code>	Method not offered by source	Planner replans

18.3 Fallback Example

```
Copernicus unavailable
→ use INCOIS if appropriate for the variable/region
→ if no authoritative substitute exists, return a PARTIAL result
→ explain the missing source in the answer + Evidence panel
NEVER fabricate missing data.
```

Fallback substitutions are always **labeled** ("value from fallback source X; primary source Y unavailable").

19. Frontend

19.1 Stack

Next.js · **TypeScript** · **Tailwind CSS** · **MapLibre GL JS** · **deck.gl** (large geospatial layers when needed) · **Recharts/ECharts** (charts) · **WebSocket/SSE** for live status updates. Map tiles: MapLibre with a permitted tile provider (no proprietary key) — or Mapbox with `MAPBOX_ACCESS_TOKEN` if selected.

19.2 UI Pages

1. **Chat** — conversational interface, inline evidence chips, language selector.
2. **Marine map** — full-screen layered map.
3. **Live conditions** — current values around a location with freshness badges.
4. **PFZ map** — advisory zones + suitability overlay.
5. **Risk panel** — hazard breakdown, risk class, rejection reasons.
6. **Route panel** — route, per-segment risk, alternatives.
7. **Source / evidence panel** — provenance for every value in the current answer.
8. **Data freshness dashboard** — per-source age vs policy.
9. **Admin / source health dashboard** — adapter status, circuit breakers, error rates (RBAC-protected).

19.3 Map Layers

PFZ · SST · chlorophyll · wave height · wind · currents · cyclone · lightning · EEZ · territorial waters · protected areas · bathymetry · route · vessel/fishing effort (where permitted by GFW license).

Each layer shows: legend, source attribution, source-update + retrieval timestamps, freshness badge.

20. Backend

Python 3.12+ · FastAPI · Pydantic v2 · httpx / asyncio · SQLAlchemy (+ asyncpg) · GeoAlchemy2 · Celery · Redis · PostgreSQL/PostGIS · MinIO/S3 · xarray · NumPy · pandas · GeoPandas · Shapely · rasterio · netCDF4 · h5py · Zarr · PyArrow · copernicusmarine · requests (simple provider clients where appropriate) · BeautifulSoup/Playwright ONLY for compliant fallback scraping (Mode D).

20.1 Internal API Contracts (representative)

Endpoint	Method	Purpose
/api/v1/chat	POST (SSE stream)	Conversational query; returns streamed answer + evidence refs
/api/v1/conditions	GET	Normalized current conditions near lat/lon (vars, radius)
/api/v1/pfz	GET	Current PFZ candidates for a sector
/api/v1/route	POST	Safest route request (origin, destination/zone, vessel class)
/api/v1/warnings	GET	Active warnings for area
/api/v1/evidence/{answer_id}	GET	Full evidence bundle for an answer
/api/v1/layers/{layer}	GET	GeoJSON/vector-tile map layer
/api/v1/admin/sources	GET/PATCH	Source registry + health (RBAC: admin)
/ws/live	WS/SSE	Warning + freshness push updates

All responses embed the canonical data schema (Sec. 11) for every value.

21. Background Workers

Worker	Cadence driver	Function
Source health worker	Per-source interval	<code>health_check()</code> all adapters; update registry + breaker state
PFZ refresh worker	Provider advisory cadence	Fetch new PFZ advisories when issued
Weather/warning refresh worker	CRITICAL policy (1–5 min)	Poll IMD warnings; push via event channel
Ocean forecast refresh worker	Forecast cycle cadence	Pull OSF/LSF/Copernicus cycles
Satellite/NRT worker	Product cadence	MOSDAC/Copernicus NRT products

Worker	Cadence driver	Function
Historical ingestion worker	On-demand/batch	ERDDAP historical subsets, archives
Geospatial version checker	Weekly/monthly	Check Marine Regions / GEBCO / WDPa versions; ingest updates
Source metadata worker	Daily	Refresh ERDDAP catalogs, dataset registry metadata
Cache cleanup worker	Hourly	Evict expired keys, compact freshness metadata

Rule: do not poll every source at the same frequency — each worker follows provider cadence + the freshness policy (Sec. 9.2).

22. Optional Event Bus

- **MVP:** FastAPI + Celery + Redis is sufficient (task queue + pub/sub). **Kafka is NOT required for the initial SIH MVP unless scale demands it.**
- **Scale path:** Kafka or Redpanda, schema registry if needed, with event topics:
`source.updated` · `warning.updated` · `pfz.updated` · `ocean.updated` ·
`weather.updated` · `satellite.updated` · `source.failed` .
- The internal publish interface is abstracted so Redis pub/sub → Kafka migration does not touch business logic.

23. Security

Area	Requirement
Secrets	Secret manager (prod); env vars (dev only); rotation; redaction (Sec. 8)
User auth	OAuth 2.0 / JWT for ORCA users; short-lived access tokens + refresh
Authorization	RBAC: <code>user</code> , <code>researcher</code> , <code>admin</code> (admin = source configuration, registry edits)
Transport	TLS everywhere (HTTPS/WSS); HSTS
At rest	Disk/database encryption where the platform supports it; encrypted object-store buckets
Audit	Audit logs for admin actions, source-config changes, credential events
API protection	Per-user and per-IP rate limiting; request size limits
SSRF	Domain allowlist for all configurable connectors ; no user-supplied URLs reach the Gateway; block private IP ranges
Outbound	Bounded timeouts, egress allowlist
Supply chain	

Area	Requirement
	Dependency scanning (pip-audit/safety), container scanning (trivy), SAST (bandit/semgrep), DAST where practical
LLM safety	Prompt-injection resistant tool design: tools validate all params; no tool executes raw URLs or SQL from model text

24. Observability

Stack: **Prometheus** · **Grafana** · **OpenTelemetry (traces)** · **structured JSON logging** · **Sentry (optional)**.

Metrics (minimum):

Metric	Type
API latency (p50/p95/p99 per endpoint)	histogram
API error rate	counter
Source freshness (age per source/dataset)	gauge
Cache hit ratio	gauge
Connector success rate	counter ratio
Data age served in answers	histogram
Query latency (end-to-end per mode)	histogram
Agent latency (per agent/step)	histogram
LLM token usage / cost	counter
Fallback events	counter
Source disagreement rate	counter

Dashboards: user-facing freshness dashboard (Sec. 19.2 #8) + ops source-health dashboard (#9).
Alerts: circuit-breaker open, warning-feed stale > policy, normalization error spike, secret-scan hit.

25. DevOps

25.1 Development

- **Docker Compose:** api, worker, beat, postgres+postgis, redis, minio, (optional) grafana/prometheus.
- **.env.example** committed (Sec. 37); **.env** git-ignored.

- **Mock source adapters** for all providers → deterministic local dev/demo without credentials.

25.2 Production

- Docker images (multi-stage, non-root); **Kubernetes optional** (Compose/managed containers acceptable for SIH scale).
- Managed Postgres / managed Redis where available; object storage (S3-compatible).
- **CI/CD with GitHub Actions**: lint → type-check → unit tests → contract tests (fixtures) → build → scan → deploy.
- Reverse proxy / API gateway (nginx/traefik) with HTTPS termination.

26. Testing

Level	Coverage
Unit	connectors · parsers · normalizers · freshness logic · spatial calculations · temporal matching · route scoring
Integration	provider mocks · ERDDAP query (recorded fixtures + optional live smoke) · Copernicus subset · IMD adapter · MOSDAC adapter · GFW adapter
End-to-end	fishing query (UC-1) · marine warning query · live ocean query · route query · ISRO event query
Failure	provider down · malformed response · stale data · conflicting sources · missing values · timeout · authentication failure · rate limiting

Rules: recorded-fixture contract tests run in CI on every commit; live smoke tests run on schedule (never hammering providers); every acceptance criterion in Sec. 35 maps to at least one automated E2E test.

27. Database Design (ERD Summary + Schemas)

27.1 Entity-Relationship Overview

```
users 1—N conversations 1—N messages
source_registry 1—N source_health
source_registry 1—N datasets 1—N dataset_variables
source_registry 1—N data_observations / forecasts / warnings / pfz_zones
recommendations 1—N evidence N—1 data_observations|forecasts|warnings
routes N—1 recommendations
marine_boundaries / protected_areas / bathymetry (static versioned layers)
audit_logs (append-only)
```

27.2 Table Definitions (PostgreSQL + PostGIS)

```
-- ===== IDENTITY & CONVERSATION =====
CREATE TABLE users (
  id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  email       TEXT UNIQUE,
  phone       TEXT UNIQUE,
  display_name TEXT,
  role        TEXT NOT NULL DEFAULT 'user',      -- user|researcher|admin
  language_pref TEXT NOT NULL DEFAULT 'en',
  created_at  TIMESTAMPTZ NOT NULL DEFAULT now(),
  last_login_at TIMESTAMPTZ
);

CREATE TABLE conversations (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  user_id UUID REFERENCES users(id),
  title TEXT, language TEXT,
  created_at TIMESTAMPTZ NOT NULL DEFAULT now(),
  updated_at TIMESTAMPTZ
);

CREATE TABLE messages (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  conversation_id UUID NOT NULL REFERENCES conversations(id),
  role TEXT NOT NULL,                        -- user|assistant|system|tool
  content TEXT NOT NULL,
  query_mode TEXT,                          -- standard|live
  answer_id UUID,                           -- links to recommendations/evidence bundle
  created_at TIMESTAMPTZ NOT NULL DEFAULT now()
);

CREATE INDEX idx_messages_conv_time ON messages(conversation_id, created_at);

-- ===== SOURCE GOVERNANCE =====
CREATE TABLE source_registry (
  source_id      TEXT PRIMARY KEY,           -- e.g. 'incois_erddap'
  provider       TEXT NOT NULL,
  name          TEXT NOT NULL,
  interface_type TEXT NOT NULL,             -- rest_api|erddap|ogc|opendap|sdk|download|
  web_fallback   TEXT,
  base_url       TEXT,
  auth_type      TEXT NOT NULL,             -- none|api_key|basic|token|account|
  unknown_pending_verification TEXT,
  credential_env_vars TEXT[],               -- names only, NEVER values
  coverage       TEXT,
  variables      TEXT[],
  update_frequency TEXT,
  expected_latency TEXT,
  freshness_policy JSONB,                  -- class, max_age_s per mode
  license        TEXT,
  attribution     TEXT,
  terms_url      TEXT,
  status         TEXT NOT NULL DEFAULT 'inactive',
  last_success    TIMESTAMPTZ,
  last_failure    TIMESTAMPTZ,
  last_source_update TIMESTAMPTZ,
  last_retrieved  TIMESTAMPTZ,
  endpoint_verified BOOLEAN NOT NULL DEFAULT false,
  notes          TEXT
);

-- Credential *metadata* only. NEVER raw secrets.
CREATE TABLE source_credentials_metadata (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),

```

```

source_id TEXT NOT NULL REFERENCES source_registry(source_id),
credential_name TEXT NOT NULL,           -- env var / secret-manager key name
requirement TEXT NOT NULL,               -- required|optional|unknown_until_verified
rotation_period_days INT,
last_rotated_at TIMESTAMPTZ,
stored_in TEXT NOT NULL DEFAULT 'secret_manager'
);

CREATE TABLE source_health (
  id BIGSERIAL PRIMARY KEY,
  source_id TEXT NOT NULL REFERENCES source_registry(source_id),
  checked_at TIMESTAMPTZ NOT NULL DEFAULT now(),
  ok BOOLEAN NOT NULL,
  latency_ms INT,
  error_code TEXT,                       -- Sec.18 taxonomy
  detail TEXT
);
CREATE INDEX idx_health_source_time ON source_health(source_id, checked_at DESC);

-- ===== DATASET REGISTRY (incl. ERDDAP registry, Sec. 5.3) =====
CREATE TABLE datasets (
  dataset_id TEXT PRIMARY KEY,           -- e.g. 'incois_oceansat2_datasets'
  source_id TEXT NOT NULL REFERENCES source_registry(source_id),
  title TEXT,
  spatial_coverage GEOMETRY(POLYGON, 4326),
  temporal_start TIMESTAMPTZ,
  temporal_end TIMESTAMPTZ,              -- last available source time
  last_retrieved TIMESTAMPTZ,
  update_frequency TEXT,
  freshness_class TEXT,                  -- critical|high|medium|low|static
  data_type_flag TEXT NOT NULL,          -- live|nrt|forecast|historical|static
  access_url TEXT,
  response_formats TEXT[],
  auth_required BOOLEAN,
  quality_metadata JSONB,
  status TEXT NOT NULL DEFAULT 'unverified'
);

CREATE TABLE dataset_variables (
  id BIGSERIAL PRIMARY KEY,
  dataset_id TEXT NOT NULL REFERENCES datasets(dataset_id),
  name TEXT NOT NULL,                    -- CHL, KD490, TSM, ...
  standard_name TEXT,
  unit TEXT,
  valid_min DOUBLE PRECISION,
  valid_max DOUBLE PRECISION,
  missing_value_encoding TEXT
);

-- ===== CANONICAL DATA (partition by time/source at scale) =====
CREATE TABLE data_observations (
  id BIGSERIAL,
  source_id TEXT NOT NULL,
  dataset_id TEXT,
  variable TEXT NOT NULL,
  value DOUBLE PRECISION,
  unit TEXT NOT NULL,
  geom GEOMETRY(POINT, 4326) NOT NULL,
  observed_at TIMESTAMPTZ,
  source_updated_at TIMESTAMPTZ,
  retrieved_at TIMESTAMPTZ NOT NULL,
  data_type TEXT NOT NULL,               -- observation|nrt

```

```

    quality_flag TEXT,
    processing_version TEXT,
    provenance_id UUID,
    confidence REAL,
    PRIMARY KEY (id, retrieved_at)
) PARTITION BY RANGE (retrieved_at);
CREATE INDEX idx_obs_var_time ON data_observations(variable, observed_at);
CREATE INDEX idx_obs_geom ON data_observations USING GIST (geom);

CREATE TABLE forecasts (
    id BIGSERIAL,
    source_id TEXT NOT NULL,
    dataset_id TEXT,
    variable TEXT NOT NULL,
    value DOUBLE PRECISION,
    unit TEXT NOT NULL,
    geom GEOMETRY(POINT, 4326) NOT NULL,
    issued_at TIMESTAMPTZ NOT NULL,           -- model cycle time
    valid_from TIMESTAMPTZ NOT NULL,
    valid_until TIMESTAMPTZ,
    retrieved_at TIMESTAMPTZ NOT NULL,
    quality_flag TEXT,
    processing_version TEXT,
    provenance_id UUID,
    confidence REAL,
    PRIMARY KEY (id, valid_from)
) PARTITION BY RANGE (valid_from);
CREATE INDEX idx_fc_var_valid ON forecasts(variable, valid_from);
CREATE INDEX idx_fc_geom ON forecasts USING GIST (geom);

CREATE TABLE warnings (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    source_id TEXT NOT NULL,
    warning_type TEXT NOT NULL,              -- cyclone|high_wave|lightning|...
    severity TEXT,
    area GEOMETRY(MULTIPOLYGON, 4326) NOT NULL,
    issued_at TIMESTAMPTZ NOT NULL,
    valid_from TIMESTAMPTZ,
    valid_until TIMESTAMPTZ,
    retrieved_at TIMESTAMPTZ NOT NULL,
    text TEXT,
    provenance_id UUID,
    active BOOLEAN NOT NULL DEFAULT true
);
CREATE INDEX idx_warn_area ON warnings USING GIST (area);
CREATE INDEX idx_warn_active ON warnings(active, valid_until);

CREATE TABLE pfz_zones (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    source_id TEXT NOT NULL,
    advisory_id TEXT,
    sector TEXT,
    geom GEOMETRY(GEOMETRY, 4326) NOT NULL,   -- line or polygon per advisory
    issued_at TIMESTAMPTZ,
    valid_until TIMESTAMPTZ,
    retrieved_at TIMESTAMPTZ NOT NULL,
    attributes JSONB,
    provenance_id UUID
);
CREATE INDEX idx_pfz_geom ON pfz_zones USING GIST (geom);

-- ===== STATIC VERSIONED LAYERS =====

```

```

CREATE TABLE marine_boundaries (
  id BIGSERIAL PRIMARY KEY,
  boundary_type TEXT NOT NULL,           -- eez|territorial_12nm|contiguous|...
  name TEXT,
  geom GEOMETRY(MULTIPOLYGON, 4326) NOT NULL,
  source_version TEXT NOT NULL,         -- e.g. 'World EEZ v12'
  ingested_at TIMESTAMPTZ NOT NULL DEFAULT now()
);
CREATE INDEX idx_bound_geom ON marine_boundaries USING GIST (geom);

CREATE TABLE protected_areas (
  id BIGSERIAL PRIMARY KEY,
  wdpa_id BIGINT,
  name TEXT,
  designation TEXT,
  geom GEOMETRY(MULTIPOLYGON, 4326) NOT NULL,
  source_version TEXT NOT NULL,
  license_note TEXT,
  ingested_at TIMESTAMPTZ NOT NULL DEFAULT now()
);
CREATE INDEX idx_pa_geom ON protected_areas USING GIST (geom);

CREATE TABLE bathymetry (
  id BIGSERIAL PRIMARY KEY,
  region TEXT NOT NULL,
  rast RASTER,                          -- or gridded points table per choice
  grid_version TEXT NOT NULL,           -- 'GEBCO_2026'
  ingested_at TIMESTAMPTZ NOT NULL DEFAULT now()
);

-- ===== ANSWERS, ROUTES, EVIDENCE, AUDIT =====
CREATE TABLE recommendations (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  message_id UUID REFERENCES messages(id),
  kind TEXT NOT NULL,                  -- fishing_zone|route|conditions|
  event_status TEXT,
  summary TEXT NOT NULL,
  risk_class TEXT,                    -- safe|caution|unsafe|no_go
  confidence REAL,
  query_mode TEXT NOT NULL,
  created_at TIMESTAMPTZ NOT NULL DEFAULT now()
);

CREATE TABLE routes (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  recommendation_id UUID REFERENCES recommendations(id),
  geom GEOMETRY(LINESTRING, 4326) NOT NULL,
  distance_m DOUBLE PRECISION,
  est_duration_s INT,
  risk_score REAL,
  segment_risks JSONB,
  constraints_applied JSONB,
  created_at TIMESTAMPTZ NOT NULL DEFAULT now()
);

CREATE TABLE evidence (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  recommendation_id UUID NOT NULL REFERENCES recommendations(id),
  claim TEXT NOT NULL,
  source_id TEXT NOT NULL,
  dataset_id TEXT,
  variable TEXT,

```

```

value_ref JSONB,                                -- canonical item snapshot
observed_at TIMESTAMPTZ,
valid_time TIMESTAMPTZ,
retrieved_at TIMESTAMPTZ,
provenance_id UUID,
processing_step TEXT,
confidence REAL
);
CREATE INDEX idx_evidence_rec ON evidence(recommendation_id);

CREATE TABLE audit_logs (
  id BIGSERIAL PRIMARY KEY,
  actor UUID,
  action TEXT NOT NULL,
  target TEXT,
  detail JSONB,
  at TIMESTAMPTZ NOT NULL DEFAULT now()
);

```

`source_credentials_metadata` stores **names, requirements and rotation metadata only** — raw secrets are NEVER stored in the database.

28. Source Registry Example

```

{
  "source_id": "incois_erddap",
  "provider": "INCOIS",
  "name": "INCOIS ERDDAP Server",
  "interface_type": "erddap",
  "base_url": "https://erddap.incois.gov.in/erddap/",
  "auth_type": "none_expected_verify_per_dataset",
  "credential_env_vars": [],
  "coverage": "Indian Ocean region (per dataset)",
  "variables": ["CHL", "KD490", "TSM", "(per registered dataset)"],
  "update_frequency": "per dataset",
  "expected_latency": "seconds (subset query)",
  "freshness_policy": {"class": "per_dataset", "note": "ERDDAP is access layer; freshness is dataset-driven"},
  "license": "per dataset – verify",
  "attribution": "INCOIS",
  "terms_url": "https://erddap.incois.gov.in/erddap/",
  "status": "active",
  "last_success": "2026-09-10T08:55:00Z",
  "last_failure": null,
  "last_source_update": "2020-05-01T00:00:00Z (incois_oceansat2_datasets – HISTORICAL)",
  "last_retrieved": "2026-09-10T08:55:00Z",
  "endpoint_verified": true,
  "notes": "Known dataset incois_oceansat2_datasets is HISTORICAL (2011-02-02..2020-05-01). NRT dataset discovery per Sec 5.5."
}

```

A second example for a discovery-phase source:

```
{
  "source_id": "incois_osf",
  "provider": "INCOIS",
  "name": "INCOIS Ocean State Forecast",
  "interface_type": "web_operational_service",
  "base_url": "https://www.incois.gov.in/oceanservices/osfforecast.jsp",
  "auth_type": "unknown_pending_verification",
  "credential_env_vars": ["INCOIS_API_KEY", "INCOIS_USERNAME", "INCOIS_PASSWORD"],
  "endpoint_verified": false,
  "status": "discovery",
  "notes": "Endpoint discovery/verification required (Sec 6.4). Production traffic blocked until verified."
}
```

29. Raw Data Provenance

For **every** downloaded or retrieved payload, the Gateway records:

Field	Description
request_id	Generated UUID per retrieval
retrieved_at	Retrieval timestamp (UTC)
source_url	Exact URL/service called
query_params	Parameters excluding secrets
checksum	SHA-256 for files
raw_payload_key	Object-store key (where storage is permitted by license)
parser_version	Version of the parsing code
normalization_version	Version of normalization rules
data_quality	QC summary
source_timestamp	Provider-declared update time

Object-store key convention:

```
raw/incois/2026/09/10/<request-id>.json
raw/copernicus/2026/09/10/<request-id>.nc
raw/mosdac/2026/09/10/<request-id>.nc
```

Provenance rows link canonical items → raw payloads → answers (evidence), completing the traceability chain (Non-negotiable #16).

30. Live Scraping Policy

Access-method hierarchy (strict order):

1. Official REST API
2. Official scientific data API/service
3. ERDDAP
4. OGC / STAC / OPeNDAP
5. Official Python/CLI client
6. Official downloadable feed/file
7. Official webpage **backend request** discovered through browser network inspection (if publicly accessible and permitted)
8. HTML/browser scraping — **last resort**
9. Third-party scraping — only if legally/technically appropriate and absolutely necessary

Scraping conduct rules: respect robots.txt where applicable · respect terms of use · respect rate limits · identify an ORCA user agent · cache results · use conditional requests (ETag/If-Modified-Since) where supported · **do not hammer websites** · **do not bypass CAPTCHA/login/access control** · monitor parser breakage · retain provenance · **alert when page structure changes**.

Scraped values are labeled with `interface_type: web_fallback` in provenance and rank lowest in source-authority weighting.

31. Important Data Policy — Data-Type Labels

ORCA strictly distinguishes and never mixes these labels:

observation · **forecast** · **nowcast** · **NRT** · **historical** · **climatology** · **static geospatial layer** · **modelled estimate**

Canonical examples:

Data	Label
Historical Oceansat-2 OCM (2011–2020)	historical
INCOIS OSF	forecast
IMD warning	warning/advisory
GEBCO	static bathymetry
GFW	modelled/derived fishing activity (depending on dataset)
OBIS	biodiversity occurrence observations
Copernicus NRT products	NRT / model / observation depending on the product

Labels are enforced at the schema level (`data_type` NOT NULL) and rendered in every UI value chip.

32. Multilingual

Initial languages: English · Hindi · Malayalam · Tamil · Telugu · Kannada · Marathi · Bengali · Odia.

Behavior requirements:

- **Detect** the user's language automatically (with manual override).
- **Answer in the user's language.**
- **Preserve location names** (no lossy transliteration of place names; show original + local script where helpful).
- Use **domain terminology consistently** (glossary-driven translation for terms like "significant wave height", "PFZ").
- **Preserve source names and technical units** (e.g., "INCOIS", "m/s") — units are never numerically altered by translation.
- Numerals and units follow the locale display convention without changing values.

Acceptance: a Malayalam UC-1 query returns a Malayalam answer with intact place names, source names, units and timestamps.

33. Performance Targets

Target	Value
Normal cached query	< 2–3 s where practical
Live multi-source query	< 5–10 s where practical
Map layer rendering	Responsive (< 1 s tile/layer swap on broadband)
Source timeout	Bounded per connector (default 10 s, warning feeds 5 s)
Connector execution	Parallel fan-out for independent sources
Cache hit ratio	≥ 70 % of STANDARD-mode reads (target)
Internal API availability	99 %+ in production where infrastructure permits

These are ORCA engineering targets. **ORCA does not promise provider availability.**

34. MVP Scope

P0 (SIH MVP — must ship)

chat · map · Planner Agent · Data Gateway · INCOIS PFZ · INCOIS OSF/LSF (after verified access) · IMD weather/warnings · ERDDAP connector · Copernicus ocean data · Marine Regions (EEZ geofencing) · GEBCO regional bathymetry · Redis · Postgres/PostGIS · evidence/provenance · freshness engine · safety/risk scoring · **one flagship fishing workflow (Sec. 14)**

P1 (enhanced)

MOSDAC · GFW · Open-Meteo supplementary weather · better multilingual · route optimization · source health dashboard · advanced anomaly analysis

P2 (advanced)

OBIS · Argo · biodiversity reasoning · advanced historical climatology · Kafka/Redpanda · advanced multimodal satellite analysis

35. Acceptance Criteria

ID	Criterion	Test
AC-A Live query	Given a current query, ORCA retrieves fresh authoritative data when cached data is stale	E2E: seed stale cache; assert forced refresh + new <code>retrieved_at</code>
AC-B ERDDAP	Given a location/time/variable request, ORCA queries the selected INCOIS ERDDAP dataset and returns only the required subset	Integration: assert subset query params + payload bounds
AC-C Freshness	Every returned data item shows source timestamp AND retrieval timestamp	E2E: schema assertion on every answer item
AC-D Safety	If a critical warning is detected, ORCA does NOT recommend a conflicting fishing zone	E2E: inject mock cyclone warning; assert candidate rejection + NO-GO
AC-E Geofencing	A candidate route intersecting a restricted/protected polygon is rejected or flagged per configured policy	E2E: route through test polygon; assert rejection/flag
AC-F Failure	If one provider fails, ORCA uses an approved fallback or clearly states unavailability	Failure test: kill mock provider; assert labeled fallback/partial answer
AC-G Provenance	Every recommendation is traceable to evidence	E2E: walk evidence → provenance → raw payload chain
	System configuration contains ONLY verified endpoints	CI check: all <code>endpoint_verified=false</code> sources blocked from prod routing; config lint

ID	Criterion	Test
AC-H No hallucinated APIs		

36. Developer Deliverables (checklist)

- ☐ Architecture diagram (Sec. 2)
- ☐ Sequence workflows (Sec. 14, 15)
- ☐ ERD / database schema (Sec. 27)
- ☐ Source integration matrix (Sec. 7)
- ☐ API contracts (Sec. 20.1)
- ☐ Data schemas (Sec. 11)
- ☐ Endpoint configuration strategy (Sec. 4.3, 5.5, 6.4, 28)
- ☐ Environment-variable template (Sec. 37)
- ☐ Queue/worker design (Sec. 21, 22)
- ☐ Caching strategy (Sec. 9, 12)
- ☐ Freshness algorithm (Sec. 9.3)
- ☐ Source health algorithm (Sec. 21 + 18)
- ☐ Error taxonomy (Sec. 18.2)
- ☐ Retry policy (Sec. 18.1)
- ☐ Security model (Sec. 23)
- ☐ Observability model (Sec. 24)
- ☐ Deployment architecture (Sec. 25)
- ☐ Test plan (Sec. 26)
- ☐

MVP roadmap + P0/P1/P2 prioritization (Sec. 34, 36.2)



Story/epic breakdown (Sec. 36.1)



Acceptance criteria (Sec. 35)



Risks and mitigations (Sec. 36.3)



Data licensing/attribution considerations (Sec. 36.4)



Operational runbook outline (Sec. 36.5)

36.1 Epic / Story Breakdown

Epic	Stories (representative)	Priority
E1 Data Gateway core	adapter base class · retry/breaker/rate-limit · provenance capture · registry CRUD	P0
E2 Endpoint discovery	INCOIS PFZ contract verification · OSF/LSF network inspection + verification · IMD endpoint confirmation · ISRO source survey	P0 (blocking)
E3 ERDDAP connector	catalog listing · subset queries · dataset registry · NRT discovery workflow	P0
E4 Copernicus connector	Toolbox auth · physics subset · waves subset · colour subset	P0/P1
E5 Static geodata	Marine Regions ingest (EEZ v12) · GEBCO_2026 regional ingest · geofencing queries	P0
E6 Freshness engine	policy config · staleness algorithm · badges API	P0
E7 Agents + planner	tool catalog · planner · PFZ/ocean/weather/geo/risk/route/evidence agents	P0
E8 Flagship workflow	UC-1 E2E, candidate filtering, ranking, route, evidence panel	P0
E9 Frontend	chat · map · layers · risk/route/evidence panels · freshness dashboard	P0
E10 Warnings pipeline	IMD warnings worker · push updates · NO-GO override	P0
E11 ISRO event demo	source monitoring · critical-mode flow (UC-6)	P1
E12 MOSDAC + GFW	download API client · effort layers (license-gated)	P1
E13 Observability + admin	metrics · dashboards · admin RBAC · source health UI	P1
E14 OBIS/Argo analytics	biodiversity/subsurface context	P2

36.2 MVP Roadmap (indicative)

Phase	Weeks	Outcome
0 — Foundations	1–2	Compose stack, registry, mock adapters, CI
1 — Discovery + connectors	2–5	E2 verification complete; ERDDAP/IMD/Copernicus connectors live
2 — Fusion + agents	4–7	Freshness engine, fusion, risk/route, planner
3 — Flagship demo	6–9	UC-1 end-to-end + evidence panel + map UI
4 — Hardening	8–10	Failure tests, observability, security review, demo polish

36.3 Risks & Mitigations

Risk	Likelihood	Impact	Mitigation
INCOIS OSF/LSF/PFZ machine endpoints not publicly available	Medium	High (PO sources)	Early discovery spike (E2); official downloadable products; compliant page parsing fallback; request INCOIS authorized access; Copernicus labeled substitute for ocean state
IMD API access constraints/keys	Medium	High	Verify early; apply for access; Open-Meteo strictly labeled supplement (never for warnings)
Provider downtime during demo	Medium	Medium	Recorded-fixture demo mode (clearly labeled), stale-cache fallback with explicit notices
WDPA license restricts deployment	Medium	Medium	License review before enabling layer; ship without it if unresolved
GFW latency misinterpreted as real-time	Low	Medium	Latency tracking + explicit labeling (Sec. 7 row 11)
Large raster ingestion overruns storage	Low	Medium	Regional subsets only; lifecycle policies
LLM hallucination of endpoints/values	Medium	High	Tool-only data access; endpoint_verified gate; evidence-mapped claims (AC-G/H)
Multilingual quality for technical marine terms	Medium	Medium	Domain glossary; human review for launch languages

36.4 Data Licensing / Attribution

- Every source's `license`, `attribution`, `terms_url` recorded in the registry and rendered in the UI layer credits.
- **Protected Planet/WDPA: license/non-commercial review is a deployment gate.**
- GFW terms compliance (token + attribution) required before enabling layers.
- GEBCO and Marine Regions attribution displayed on bathymetry/boundary layers.
- OBIS/Argo citation practices followed for analytics outputs.

36.5 Operational Runbook (outline)

1. Deploy/rollback procedures (CI/CD, image tags).
 2. Secret rotation procedure per provider.
 3. Source outage playbook (breaker status → fallback verification → user notice).
 4. Warning-feed staleness incident procedure (highest severity).
 5. ERDDAP dataset registry maintenance (Sec. 5.5 checklist).
 6. Static layer version upgrade (Marine Regions/GEBCO/WDPA) procedure.
 7. Parser-breakage response for web_fallback sources.
 8. Capacity review (partitions, object-store lifecycle) schedule.
-

37. Required Environment File Template

`.env.example` (placeholders only — committed; `.env` is git-ignored):

```
APP_ENV=development
APP_SECRET_KEY=CHANGE_ME
JWT_SECRET_KEY=CHANGE_ME
DATABASE_URL=postgresql+asyncpg://user:password@localhost:5432/orca
REDIS_URL=redis://localhost:6379/0
MINIO_ENDPOINT=http://localhost:9000
MINIO_ACCESS_KEY=CHANGE_ME
MINIO_SECRET_KEY=CHANGE_ME
OPENAI_API_KEY=CHANGE_ME
INCOIS_API_KEY=OPTIONAL_OR_VERIFY
INCOIS_USERNAME=OPTIONAL_OR_VERIFY
INCOIS_PASSWORD=OPTIONAL_OR_VERIFY
IMD_API_KEY=OPTIONAL_OR_VERIFY
COPERNICUSMARINE_SERVICE_USERNAME=CHANGE_ME
COPERNICUSMARINE_SERVICE_PASSWORD=CHANGE_ME
MOSDAC_USERNAME=CHANGE_ME
MOSDAC_PASSWORD=CHANGE_ME
GFW_API_TOKEN=CHANGE_ME
PROTECTED_PLANET_API_TOKEN=CHANGE_ME
OBIS_API_KEY=OPTIONAL_OR_VERIFY
ISRO_API_KEY=OPTIONAL_OR_VERIFY
MAPBOX_ACCESS_TOKEN=OPTIONAL
SENTRY_DSN=OPTIONAL
```

Never put real credentials in this PRD, in Git, or in any artifact.

38. Source Links (verify before final PRD publication)

Source	Official reference	Why it matters
INCOIS OSF	https://www.incois.gov.in/oceanservices/osfforecast.jsp	Operational ocean forecast variables and current interface
INCOIS LSF	https://incois.gov.in/oceanservices/LSF/index.html	Location-specific wave/current forecast interface
INCOIS Data Holdings	https://incois.gov.in/site/dataholdings.jsp	Catalog of INCOIS data products
INCOIS ERDDAP	https://erddap.incois.gov.in/erddap/	Programmatic scientific data access
INCOIS ERDDAP status	https://erddap.incois.gov.in/erddap/status.html	Server status
IMD API Reference	https://api.imd.gov.in/public/api_reference.html	Weather/warning API reference
Copernicus Marine Toolbox	https://help.marine.copernicus.eu/en/articles/7949409-copernicus-marine-toolbox-introduction	Official Python/CLI programmatic access
Copernicus Subset API	https://help.marine.copernicus.eu/en/articles/8283072-copernicus-marine-toolbox-api-subset	Targeted subsetting
MOSDAC API manual	https://www.mosdac.gov.in/downloadapi-manual	Satellite data download API; NRT/archive support
Global Fishing Watch API	https://globalfishingwatch.org/our-apis/documentation/	REST APIs and token authentication
OBIS data access	https://portal.obis.org/data/access/	API/GeoParquet access options
GEBCO 2026 grid	https://www.gebco.net/data-products-gridded-bathymetry-data/gebco2026-grid	Current 2026 bathymetry grid
Marine Regions	https://www.marineregions.org/downloads.php	EEZ and maritime boundary downloads/services
Protected Planet API	https://api.protectedplanet.net/documentation	Current API v4 documentation and token requirement
Open-Meteo	https://open-meteo.com/	Supplementary weather source
Argo	https://www.argodatamgt.org/DataAccess.html	Programmatic Argo data access
ISRO	https://www.isro.gov.in/	Official mission information source

39. Final Statement of Compliance

This PRD:

- distinguishes **confirmed provider capabilities** from assumptions (⚠ markers, `endpoint_verified` gates);
- **never invents endpoints or API keys** and never exposes secrets;
- explains ERDDAP, ORCA ↔ ERDDAP connectivity, operational-service connectivity, API-first live acquisition, fallback scraping, freshness, historical-vs-live separation, provenance, safety handling, source failure, authentication, rate limiting, caching, spatial/temporal alignment, agent workflows, tech stack, deployment, testing, security, and monitoring;
- states, canonically: **"ORCA continuously acquires live data through the best available authoritative machine-readable interface, with API/service access first and compliant scraping only as a last resort."** — never "ORCA scrapes all websites continuously";
- treats the **210 MB Oceansat-2 OCM dataset as historical/reference data**, never as today's operational ocean condition source;
- is suitable for **SIH 2026 development, technical review, developer handoff, architecture review, demo preparation, and future production scaling.**

— End of PRD —